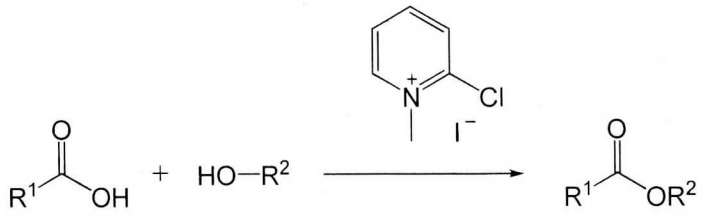
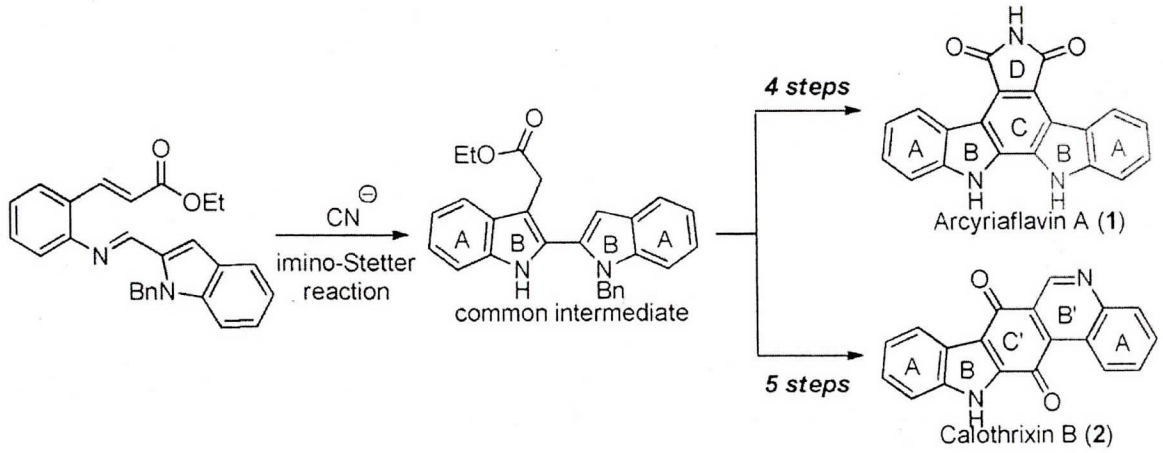
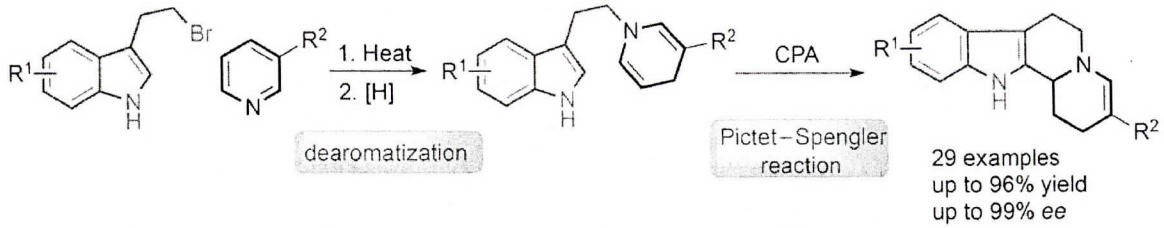
Organic Cumulative Exam
November 10, 2017

1. (20 points) Write a mechanism for the following esterification reaction.



2. (30 points) Write a mechanism for the first step imino-Stetter Reaction (*Org. Lett.*, **2017**, 19 (11), pp 2785–2788, DOI: 10.1021/acs.orglett.7b00687).



3. (50 points) Write a mechanism for the last step in the following figure. Simply treat CPA as a strong Brønsted acid H^+. (*Angew. Chem. Int. Ed.* **2017**, *56*, 7440. DOI: 10.1002/anie.201703178)